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Causal discovery by continuous optimization with weighted superstructure

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ABSTRACT

Recently, score-based methods that frame causal discovery as a continuous optimization problem have achieved significant performance gains. However, their performance often degrades in scenarios characterized by high dimensionality, limited samples, and heterogeneous noise. It has been empirically observed that constraint-based methods generally perform better in learning causal structures under heterogeneous noise with fewer samples than the continuous optimization based methods, which motivates us to improve continuous optimization based causal discovery by introducing and exploiting conditional independence (CI) information. To obtain reliable CI information, we use low-order (0-order and 1-order) CI tests to construct a weighted superstructure from the observed data. Then, we propose the weighted CI constraints into continuous optimization, where the constraints can be easily obtained from the weighted superstructure. Theoretically, we provide a convergence guarantee for our constrained optimization framework with the CI constraints. Extensive experiments on synthetic and real-world datasets demonstrate that our proposal effectively improves the performance of existing continuous optimization methods in causal discovery, particularly in the low-sample regime. The source code and data are available at https://github.com/Mjchen22/WIC.

1. Introduction

Structure learning of Directed Acyclic Graphs (DAGs) from observational data is a fundamental problem in artificial intelligence (AI) and it is important for some research areas such as causal inference (Feder et al., 2022), recommender systems (Gao et al., 2024) and bioinformatics (He et al., 2025; Pearl & Mackenzie, 2018). In causal structure learning, independence and conditional independence (CI) tests provide a flexible approach to assess the CI relationship between variables. Under the faithfulness assumption, these tests allow researchers to utilize CI tests to learn graphical models that may have causal interpretability, such as the Markov Equivalence Class (MEC) (Ogarrio et al., 2016). Methodologies for causal structure learning are primarily divided into two categories: constraint-based and score-based. Constraint-based methods, such as the PC algorithm (Spirtes & Glymour, 1991), typically obtain a partial DAG (Pearl, 2009) by applying CI tests, thereby modeling the relationships between variables more concisely. In contrast, score-based methods, like Greedy Equivalence Search (GES) (Chickering, 2003), define a global score function and search for a graph structure that best fits the data, thus avoiding the need for individual CI tests (Solus et al., 2021; Zhang et al., 2023). However, the combinatorial nature of the graph search space means that many score-based methods face significant computational challenges, especially in high-dimensional data.

Recently, to reduce the complexity of graph combinatorial search and enhance the performance of causal discovery in high-dimensional data, researchers have increasingly tried to frame causal discovery as a continuous optimization problem (Ng et al., 2020). NOTEARS (Zheng et al., 2018) introduces a novel differentiable characterization of graph acyclicity. By formulating this property as a constraint within a standard score-based objective, the entire discrete search problem is transformed into a continuous one that can be solved efficiently with gradient-based optimization. Originally designed for linear DAG models, NOTEARS has since been extended to various complex scenarios, including non-linear and non-parametric models, using neural networks (Brouillard et al., 2020; Ng et al., 2020; Pamfil et al., 2020; Sun et al., 2023;

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Wei et al., 2020; Yu et al., 2019; Zhang et al., 2022; Zheng et al., 2020). Subsequent works have further refined this paradigm, including NTS-NOTEARS (Sun et al., 2023), which leverages time series data to learn non-parametric DAGs, and DAGMA (Bello et al., 2022) that is based on the log-determinant function. These approaches are commonly referred to as continuous optimization methods or differentiable score-based methods.

Although these methods achieve impressive performance, they rely on strong assumptions about the parametric family of models and the distributions of variables, which are necessary to ensure that the causal structure is fully or partially identifiable (Ng et al., 2022b; Peters et al., 2014). More importantly, the objective functions of existing methods are constrained by various factors to recover as many edges as possible and approximate the ground truth causal graph. However, during the optimization process, this constraint often leads the optimization process to converge on graphs with false edges, especially in the case of fewer samples. Neglecting the reduction of false edges can lead to early-stage errors that are irreparable in later-stage, ultimately resulting in poor robustness.

To address these challenges, we propose to leverage the weighted superstructure w.r.t. the observations as a constraint in continuous optimization for learning DAGs. Specifically, under the faithfulness assumption (Ramsey et al., 2006), if variables x and y are conditionally independent given a set Z (i.e., $x \perp\!\!\!\perp y \mid Z$), we can conclude that there is no direct causal relationship between x and y, that is, there should be no direct edge between them. This allows us to use the CI relationship between x and y to constrain the edge between them. Notably, CI tests are prone to failed detections of true edges between variables with a larger conditional set, leading to Type II errors.

To get the CI information as much reliable as possible, we construct a weighted superstructure w.r.t. the observations by low-order CI tests. This weighted superstructure, represented as a CI matrix M, which encodes the confidence of the existence of edges between any two nodes, is then utilized as prior knowledge in the continuous optimization task. This approach guides the optimization process to eliminate edges between pairs of variables that are likely independent, while still granting the combinatorial search flexibility to discover the complete graph structure. On the other hand, assuming access to a perfect CI oracle, we can derive a set of constraints for the continuous optimization task, on the order of $O((d-k)*d)$ constraints for a true causal graph with d variables and average degree k. Generally, $k \ll d$, the CI oracle provides CI information and the weighted superstructure, i.e., the CI matrix M represents the strength of the causal relationship between variables. Therefore, the CI relationships contain rich information about the elements of M, which facilitates the convergence of the continuous optimization process to a more effective final solution. Furthermore, unlike continuous optimization methods that require strong parametric assumptions, our approach builds upon non-parametric CI tests, making it inherently more robust and applicable to complex scenarios involving nonlinear or heterogeneous data (Huang et al., 2017; Ren et al., 2025; Zhang et al., 2017b).

In this paper, we separately design hard and soft constraints for the continuous optimization process. The hard constraints leverage the weighted superstructure to ensure that certain non-causal edges in the causal graph, identified during optimization, are eliminated. Concretely, we first construct a CI matrix M of the edges by low-order CI tests on observed data. Then, we embed M into the continuous optimization equation. When the optimization procedure terminates and converges to a feasible solution, we obtain a final causal graph from which non-causal edges have been directly pruned. We refer to this hard constraint as the **W**eighted **C**onditional **I**ndependence **C**onstraint (WIC). It's important to note that, as a hard constraint, WIC may erroneously eliminate true edges if M is not estimated with complete accuracy. To address this issue, we further propose a flexible soft constraint that transforms the CI information as a regularization term in the loss function. This soft constraint is termed as **W**eighted **C**onditional **I**ndependence **R**egularization (WIR). Most existing continuous optimization methods learn a sparse DAG by incorporating L1 regularization, which imposes a uniform sparsity constraint on the causal graph during learning. However, the sparsity of different DAGs can vary significantly, and an inappropriate choice of the L1 penalty coefficient λ_1 may disrupt the balance between sparsity and accuracy in the learned causal structure, ultimately leading to suboptimal performance. In contrast to L1 regularization, WIR is a more targeted and effective technique because it selectively penalizes only the edges identified as non-causal by CI tests, thus preventing false positives without compromising the overall learned structure.

In summary, our contributions are as follows:

- We propose a novel weighted CI constraint with hard and soft versions to causal discovery based on continuous optimization, which in a sense serves as the pre-learned prior knowledge to continuous optimization and therefore effectively improves their performance. The key advantages of our method include that the weighted CI constraint can be easily calculated by low-order CI tests, and the proposed constraints can be used in most existing continuous optimization-based causal discovery methods.
- We prove the convergence guarantee of WIC under some mild conditions and the consistency of some existing methods with WIR. We further evaluate the performance of these methods under finite sample conditions, conducting a thorough analysis of the stability and computational complexity of our approach.
- We conduct extensive experiments on synthetic and real-world datasets to evaluate the proposed method, which shows that compared with existing methods, the proposed method can effectively improve causal discovery performance in both linear and non-linear scenarios, especially when the sample size is small.

The rest of this paper is organized as follows: Section 2 reviews related work and highlights the differences between our method and the existing works. Section 3 introduces the theoretical basis, technical details and the algorithmic framework of the proposed method. Section 4 provides the experiment evaluation. Section 5 offers some discussions on results, and finally Section 6 concludes this paper.

2. Related work

Extracting and inferring causal relationships from observational data is critically important across academic disciplines such as medicine, finance, biology, and machine learning. In these fields, Causal structure or graph models are commonly used to describe the joint distribution of variables in observational data and to infer valid causal relationships, with the Directed Acyclic Graph (DAG) commonly employed to represent the causal structure. DAG comprises a series of nodes that represent variables and directed edges that represent causal relationships between different variables. Causal discovery mainly intends to recover potential causal structures and associated conditional probability distributions. Most existing methods for causal structure learning from observational data are often divided into two groups: constraint-based methods and score-based methods.

2.1. Constraint-based methods

Constraint-based methods rely on statistical tests of conditional independence and are easy to understand and widely used. The general steps of constraint-based methods are to first build a skeleton between variables based on conditional independence, and then orient the skeleton according to rules such as consistent propagation. The goal is to construct Completed Partially Directed Acyclic Graphs (CPDAGs) representing the MEC of the true causal graph (Ng et al., 2020). For example, the IC (Pearl, 2009) algorithm looks for the conditional separation sets of each pair of variables in the variable set, and if the pair of variables does not have a separated set, the edge between the pair of variables is retained. Then, a partial undirected graph is obtained after direction judgments. However, this method requires calculating the independence between all nodes, which will generate lots of redundant tests and consume lots of time, especially when the number of nodes is high. PC (Spirtes & Glymour, 1991) algorithm is the most classic constraint-based causal discovery algorithm with the basic idea of iteratively removing non-causal edges in the graph by conditional independence tests. As a reliable causal discovery method, PC can handle various types of data distribution and causality. FCI (Colombo et al., 2012) can generate asymptotically correct results even in the presence of confounding factors. The CD-NOD (Zhang et al., 2017b) algorithm leverages heterogeneous data and distribution shifts to infer causal direction under more complex conditions. Recent advances also include approaches such as (Akbari et al., 2021; Li et al., 2025; Mokhtarian et al., 2021, 2025), which improve learning efficiency. However, constraint-based methods face inherent limitations in two respects: theoretically, they strictly depend on the faithfulness assumption; practically, they rely on CI tests that demand large sample sizes and are computationally expensive.

2.2. Score/optimization-based methods

Score-based methods operate on the principle of defining a global goodness-of-fit measure to evaluate candidate graph structures. This reframes causal discovery as a search for the highest-scoring graph, relaxing the strict faithfulness assumption required by individual CI tests. Consequently, the main idea of score-based methods is to assign a score to each potential causal graph and then search for the optimal one. For example, GES (Chickering, 2003) typically assumes a linear model with Gaussian noise and performs a greedy search that aims to find a high-scoring DAG in the structure space. GES-GS (Huang et al., 2018) allows capturing nonlinear dependencies between variables in causal structure learning, thereby enhancing the model's applicability and predictive capabilities. CAM (Bühlmann et al., 2014) decouples the search for the optimal node ordering from the selection of parent nodes for each individual node and assumes that the nonlinear function follows an Additive Noise Model (ANM). For large-scale graphs, optimized combinatorial search algorithms like fGES (Ramsey et al., 2017) and BOSS (Andrews et al., 2023), as well as causal partitioning techniques via edge cutting (Zhang et al., 2024), have been developed to handle high-dimensional data efficiently. However, the practical application of these methods is often limited by the computational challenge of searching a vast structure space, which becomes intractable as the number of variables grows, even with large sample sizes.

To tackle this challenge, several methods reformulate score-based learning as a continuous constrained optimization problem, allowing the use of efficient gradient-based algorithms to find a solution. NOTEARS (Zheng et al., 2018) introduces a constraint that is computationally efficient and differentiable, ensuring acyclicity in the causal graph. It formulates score-based structure learning of linear Structural Equation Models (SEMs) as a continuous optimization problem. DAG-GNN (Yu et al., 2019) extends SEM structure learning to nonlinear scenarios by modeling the causal relationships with Graph Neural Networks. GAE (Ng et al., 2019) generalizes the gradient-based method to the graph autoencoder framework, significantly improving model performance on large causal graphs. However, a common limitation of these extensions is the assumption of a uniform nonlinear transformation for all variables, which may be restrictive in practice. In particular, when the relationships between variables exhibit high heterogeneity, a uniform nonlinear transformation may be insufficient to capture the full complexity of the data. Therefore, some methods focus on designing more sophisticated parameterizations of the adjacency matrix or the score function itself. For example, GraN-DAG (Lachapelle et al., 2020) estimates an equivalent adjacency matrix and then optimizes the adjacency matrix through a pruning process. MCSL-MLP (Ng et al., 2022b) adds an augmented term based on a binary adjacency matrix and employs Gumbel-Softmax to approximate the adjacency matrix. GOLEM (Ng et al., 2020) analyzes the asymptotic role of sparsity and DAG constraints, and formulates a likelihood-based score function.

Note that the method we propose in this work is extended from the CIC/CIR method (Xia et al., 2023), a classic method that combines continuous optimization with CI information. The crucial difference between the two methods is that CIC/CIR employs a $(0,1)$-CI matrix, treating all constraints as equally certain and thus failing to exploit the varying confidence levels inherent in statistical tests. As we demonstrate in Sections 4.4 & 4.5, our new method - which utilizes a more nuanced weighted representation of CI information - significantly outperforms CIC/CIR.

3. Method

In this section, we firstly give the problem statement of the causal discovery problem (Section 3.1). Next, we propose a weighted superstructure based on CI tests and transform this superstructure into hard structural constraints within the continuous optimization problem (Sections 3.2 & 3.3). Furthermore, we introduce a method for converting these hard constraints into soft constraints (Section 3.4). We then provide the causal discovery algorithm utilizing the WIC/WIR constraints (Section 3.5). Finally, we present the stability analysis and time complexity of the proposed method (Section 3.6).

3.1. Problem statement

Given a set of random variables $X = (x_1, \dots, x_d)$ independently and identically sampled from the joint distribution of a d-dimensional $P(X)$. Our goal is to find causal relationships (a probabilistic directed graphical model) among these variables, which are encoded as a DAG $G = (V, E)$. In this graph, each node $v_i \in V$ corresponds to a variable x_i, $E \subseteq V \times V$ is the set of directed edges, where an edge $(v_i, v_j) \in E$ means that v_i is a direct cause of v_j. We assume that the probabilistic directed graphical model is parameterized by a weight matrix $W \in \mathbb{R}^{(d\times d)}$, such that the sparse pattern of W corresponds to the adjacency pattern of the graph, and satisfies $W_{ij} \neq 0$ if and only if $(i,j) \in E$. In other words, this setting represents each edge in the DAG is associated with a single parameter W_{ij}. In this paper, we assume that X is generated by a Generalized Linear structural equation Model (GLM), formally,

$$\mathbb{E}[x_j|X] = g(W_{.j}^T X), \tag{1}$$

where $W_{.j}$ is the jth column of W, g is a non-linear function.

3.2. Weighted superstructure

Consider the following score-based combinatorial optimization problem, which aims to optimize the following score function:

$$\min_{W} \; S(W;\phi) = \mathcal{L}_{rec}(W; X, \phi) + \lambda_1 \mathcal{L}_{sparse}(W), \tag{2}$$
$$\text{s.t. } \mathcal{L}_{DAG}(W) = 0,$$

where $\mathcal{L}_{rec}$ is the Maximum Likelihood Estimator (MLE), which calculates the reconstruction error with the true graph; $\mathcal{L}_{sparse}$ is the sparsity loss function, which encourages fewer edges in the graph; and $\mathcal{L}_{DAG}(W)$ is the acyclicity loss function, which is a penalty term encouraging acyclicity on W. The parameters ϕ represent additional optimizable model parameters, which are optimized alongside W. λ_1 is a hyperparameter of the sparsity penalty coefficient, which can be selected via cross-validation in practice.

To facilitate further research on the connection between CI relationships and edges in continuous optimization problems, we introduce the adjacency-faithfulness assumption (Ramsey et al., 2006). This assumption serves as a relaxation of the standard faithfulness condition (Spirtes et al., 2001). Unlike standard faithfulness, which requires a strict equivalence between all statistical independencies and d-separations, adjacency-faithfulness specifically posits that adjacent variables are conditionally dependent given any subset of other variables. This assumption is sufficient for our framework and provides the theoretical justification for pruning edges via CI tests. Under this assumption, we show that for general structures, if the ground truth DAG is acyclic, adding CI structural constraints can reduce the uncertainty of W and guarantee learning a DAG similar to the ground truth DAG.

Assumption 1. (Adjacency-faithfulness). Let G be the true causal DAG over a set of variables X. For any pair of variables (x_i, x_j) that are adjacent in G, they are conditionally dependent given any other subset of variables $S \subseteq X\backslash_{x_i,x_j}$.

Assumption 1 implies that if there is no edge directly connecting two variables in graph G, then the two variables should be independent in the distribution $\mathbb{P}$. Therefore, when given the other variables, CI in $\mathbb{P}$ implies nonadjacency in G. If the data is generated by GLM Eq. (1), the existence of edges in the graph can be represented by the weight matrix W. Then we can draw the connection between edges and CI, and treat CI as prior knowledge in continuous optimization. Specifically, while continuous optimization evaluates the entire weight matrix W, CI testing can be used to directly identify its zero-valued elements. Based on this connection, the CI matrix related to the observed variables is defined as:

Definition 1. (CI matrix). Given a causal graph (a DAG) $G = (V, E)$, for any $i, j \in V$, if they are non-adjacent in G and are D-separated by a set $Z \subset V\backslash_{x_i,x_j}$, we say that x_i and x_j are conditionally independent given Z. The CI relationship can be represented as a matrix, and if two variables are independent given other variables, then the corresponding elements in the CI matrix will have specific values. The CI matrix M^{bin} is calculated as follows:

$$M^{bin}_{i,j} \triangleq \begin{cases} 1, & x \perp y|Z \\ 0, & x \not\perp y|Z \end{cases} \tag{3}$$

we note that it is difficult to accurately estimate $x \perp y$ given all possible $Z \subset V\backslash_{x_i,x_j}$. Alternatively, we use only 0-order and 1-order CI tests to avoid a high Type-II error rate (Zhang et al., 2017a). As a result, we obtain a superstructure of the causal skeleton rather than the exact causal skeleton, since some spurious edges cannot be eliminated by low-order tests. We give the definition of the weighted superstructure below:

Definition 2. (Weighted superstructure) *Let $G = (V, E)$ be the true causal graph (a DAG) of the observed data D, we say an undirected weighted graph $G_{\mathcal{W}} = (V, E_{\mathcal{W}})$ is a superstructure of G based on D, if for any $i, j, e< i, j >\in E \Rightarrow e(i, j) \in E_{\mathcal{W}}$.*

In practice, the weighted superstructure corresponds to a **weighted CI matrix**, which can be efficiently calculated by the classic linear partial correlation test, as follows:

$$M_{i,j} \triangleq \begin{cases} p, & \text{if } 1 - \Phi(x) > \frac{\alpha}{2} \\ 0, & \text{if } 1 - \Phi(x) \le \frac{\alpha}{2} \end{cases} \tag{4}$$

where $\Phi(\cdot)$ denotes the cumulative distribution function (CDF) of the standard Gaussian distribution, α represents the significance level in a two-tailed test used to determine whether the variables exhibit an independent relationship, p denotes the p-value, which is derived from the CDF of the observed variable. To introduce more causal information related to the observed variables, the estimated p-value is as follows (Shi et al., 2018):

$$\Phi(x) = \Phi(\sqrt{n-3} \cdot |0.5 \cdot \ln\left(\frac{1+\hat{\rho}_{ij}}{1-\hat{\rho}_{ij}}\right)|), \tag{5}$$

$$p = 2 \cdot (1 - \Phi(x)). \tag{6}$$

The steps for calculating the p-value can be summarized as follows.

1) Randomly extract two samples x_i and x_j from the set X, and combine them with a set Z that provides additional sample information to calculate the correlation coefficient $\hat{\rho}_{ij}$ between x_i and x_j.
2) Determine the corresponding test statistic based on the calculated correlation coefficient $\hat{\rho}_{ij}$. Here, n represents the number of observations used to calculate the correlation coefficient $\hat{\rho}_{ij}$ between the two samples x_i and x_j.
3) Compare the CDF of the test statistic with the pre-set significance level α to output the result of the conditional independence test. This result splits the traditional binary output (0 or 1) and presents it as a p-value. The p-value provides more detailed information and can better represent the causal relationship between variables.

Algorithm 1 Construction of weighted CI matrix M.

Require: Observed data X, significance level α, maximum order k_{CI}.
Ensure: Weighted CI matrix M.

```
Initialize M as a zero matrix of size d × d
for each pair (v_i, v_j) ∈ V × V do
    p_best ← 0
    for each conditional set Z ⊆ Neighbors(v_i, v_j) with |Z| ≤ k_CI do
        Calculate partial correlation ρ̂_{ij|Z} from X
        Compute z-score: z = 1/2 ln((1+ρ̂)/(1−ρ̂)) · sqrt(n − |Z| − 3)
        Compute p-value: p = 2 · (1 − Φ(|z|))
        if p > α then
            p_best ← max(p_best, p).
        end if
    end for
    M_{i,j} = M_{j,i} = p_best
end for
return M
```

For clarity, the complete procedure for computing the weighted CI matrix M is summarized in Algorithm 1. Note that the CI matrix is dynamically updated according to the selected test order. This means that different orders of CI tests can significantly influence the initialization process of the CI matrix. Here, we suggest using only low-order (0-order and 1-order) CI test, because this can not only improve the accuracy of the test but also reduce its time complexity. Specifically, when performing 1-order tests, we first identify all the variable pairs found to be dependent in the 0-order tests. Then, we use an exhaustive method to determine if a variable satisfies the conditional set Z, thereby rendering these variable pairs independent under the specified condition Z. If independence is confirmed, the corresponding entry in the CI matrix is updated from zero to the corresponding p-value. This method optimizes the initialization process of the CI matrix and ensures high efficiency and accuracy when dealing with smaller sample sizes.

3.3. Weighted CI constraint

We then embed the weighted CI matrix as a hard constraint into the score function of Eq. (2), thus we have

$$\begin{aligned} \min_{W} \quad & \mathcal{S}(W; \phi) = \mathcal{L}_{rec}(W; X, \phi) + \lambda_1 \mathcal{L}_{sparse}(W), \\ \text{s.t.} \quad & \mathcal{L}_{DAG}(W) = 0, \\ & M_{i,j} \circ W_{i,j} = 0, \forall i, j \in [d]. \end{aligned} \tag{7}$$

The hard constraint in Eq. (7), termed **W**eighted Conditional **I**ndependence **C**onstraint (WIC), enforces $W_{i,j} = 0$ whenever $M_{i,j}$ is nonzero. This ensures that no direct edge exists between x_i and x_j if they are identified as conditionally independent. Finding a global optimum for this constrained problem is challenging due to the non-convexity of the acyclicity term $\mathcal{L}_{DAG}(W)$ (Nocedal & Wright, 1999). Standard optimizers often converge to stationary points, making it difficult to guarantee a feasible DAG solution. To address this, we employ the Quadratic Penalty Method (QPM) (Ng et al., 2022a), which transforms the hard-constrained problem into a sequence of unconstrained subproblems via a penalty term.

Formally, WIC functions as a weighted penalty term, formulated as $\frac{\rho}{2}\|M \circ W\|_F^2$. Unlike binary constraints that impose a uniform penalty on all identified non-causal edges, the gradient contribution of the WIC penalty is derived as $\rho M_{ij}^2 W_{ij}$. In binary methods (e.g., CIC), the term M_{ij}^2 imposes either zero or a full penalty ρW_{ij}. This binary switching makes the optimization sensitive to threshold selection and Type II errors in CI tests.

In contrast, WIC utilizes M_{ij}^2 as a continuous scaling factor. For edges with weaker independence (smaller M_{ij}), the initial penalty is correspondingly low. This mechanism prevents the optimization from prematurely discarding potentially true causal edges based on CI tests and ensures they are removed only if they prove to be truly independent as ρ increases. This dynamic weighting offers superior robustness to heterogeneous noise compared to the rigid binary constraints of CIC.

We now demonstrate the theoretical convergence of WIC and show that the causal graph can be inferred by solving Eq. (7). Our proof builds upon the class of DAG-constrained functions proposed in the NO-FEARS framework (Wei et al., 2020). We first review this constraint and then show that adding WIC maintains the original convergence guarantees. Generally, NO-FEARS defines a class of functions corresponding to matrix polynomials with positive coefficients ($c_p > 0$ for $p = 1, 2, \ldots, d$):

$$\mathcal{L}_{DAG}(W) = \sum_{p=1}^{d} C_p tr((W \circ W)^p), \tag{8}$$

any function in Eq. (8) can be used to characterize acyclicity. To learn DAG structures in continuous optimization, it is necessary to use acyclic characterization functions to ensure that the learned graph structure remains acyclic during the optimization process. The commonly used DAG acyclicity characterizations, such as $\mathcal{L}_{DAG}(W) = tr(e^{(W \circ W)}) - d$ and $\mathcal{L}_{DAG}(W) = tr((I + \mu W \circ W)^d) - d$, can both be regarded as a special case in Eq. (8). Our convergence proof for WIC follows the framework established in Ng et al. (2022a). To facilitate a clear understanding, we first outline the key assumptions and lemmas from this prior work:

Assumption 2. For a function $\mathcal{L}_{DAG}(W) \geq 0$ such that $\mathcal{L}_{DAG}(W) = 0$ if and only if its gradient $\nabla \mathcal{L}_{DAG}(W) = 0$.

Lemma 1 (Theorem in Ng (Ng et al., 2022a)). *When applying the QPM algorithm to solve Eq. (2), let the sequence of iterates be $\{W^k\}$. If the penalty coefficients diverge to infinity ($\rho_k \to \infty$), the non-negative tolerance sequence $\{\tau_k\}$ is bounded, and Assumption 2 holds for the function $\mathcal{L}_{DAG}(W)$, then any limit point W^* of the sequence $\{W^k\}$ will be feasible. In this optimization problem, feasibility implies that W^* strictly satisfies the acyclicity constraint (i.e., $\mathcal{L}_{DAG}(W) = 0$) and any other imposed hard constraints, ensuring the output is a valid DAG structure.*

Building on this foundation, we introduce Lemma 2 to establish that every function within the class defined by Eq. (8) satisfies Assumption 2. This lemma demonstrates that any function from this class can be used as a suitable DAG constraint, allowing the QPM algorithm to converge to a feasible solution under mild conditions.

Lemma 2. *Any function $\mathcal{L}_{DAG}(W)$ defined in Eq. (8) satisfies Assumption 2.*

The proof of Lemma 2 is presented in Appendix A. Lemma 2 demonstrates that our chosen DAG constraint functions satisfy the convergence conditions required by Lemma 1, thereby guaranteeing the convergence of the QPM algorithm. This allows us to extend this convergence guarantee to our WIC constrained method, which is formally stated in Theorem 1.

Theorem 1. *Consider the WIC constrained optimization problem in Eq. (7), solved via the QPM. Let the DAG constraint be any function from the class defined in Eq. (8) and the sequence of iterates be $\{W^k\}$. If the penalty coefficients diverge to infinity ($\rho_k \to \infty$), the non-negative tolerance sequence $\{\tau_k\}$ is bounded, and Assumption 2 holds for the function $\mathcal{L}_{DAG}(W)$, then any limit point W^* of the sequence $\{W^k\}$ will be feasible, corresponding to a feasible DAG structure.*

The proof of Theorem 1 is presented in Appendix B. Theorem 1 implies that if Assumption 2 is satisfied, then a feasible solution can be reached under mild conditions by combining a suitable DAG constraint with the WIC constraint.

3.4. Weighted CI regularization

However, any CI testing method on finite samples is inevitably susceptible to Type II errors, which poses a fundamental challenge. This implies that the proposed WIC hard constraints may incorrectly remove true edges, leading to poor performance due to an increased Type II error rate. As a more robust alternative, we reformulate this constraint into a soft regularization term. Concretely, we transform the hard constraint into a more flexible regularization term and introduce a regularization parameter. By adjusting the regularization parameter, we can enhance the model's fit to the data and tend the model to true causal structures rather than noise or spurious correlations. We incorporate this soft constraint into the loss function from Eq. (2) to define our new optimization problem:

$$\begin{aligned} \min_W \quad S(W;\phi) =& \mathcal{L}_{rec}(W; X, \phi) + \lambda_1 \mathcal{L}_{sparse}(W) \\ &+ \lambda_{CI} \|M \circ W\|_F^2, \\ \text{s.t.} \quad \mathcal{L}_{DAG}(W) =& 0. \end{aligned} \tag{9}$$

The soft constraint in Eq. (9) is called **W**eighted Conditional **I**ndependence **R**egularization (WIR) and introduces a regularization term $\|M \circ W\|_F^2$. This term is designed to penalize the weights influenced by M, thereby implicitly restricting the parameter space of W. In this way, the WIR encourages model sparsity and helps distinguish zero and non-zero elements in the weight matrix W.

Given that WIR incorporates a regularization term within the objective function, it is crucial to address concerns regarding its consistency. Specifically, we must determine whether this regularization term affects the objective function and whether incorporating WIR can asymptotically learn a DAG that is equivalent to the ground truth DAG. Ng et al. (2020) proposed GOLEM, which employs CI regularization. Therefore, here we first analyze the theoretical guarantee of asymptotic consistency in GOLEM, which mentions that the sparsity penalty is sufficient to asymptotically learn a DAG equivalent to the ground truth DAG. We then prove that the role of WIR is equivalent to the sparsity penalty $\mathcal{L}_{sparse}(W)$, thereby establishing that the addition of WIR imposes only a sparsity penalty on the learned graph without affecting the asymptotic consistency.

Definition 3. (Quasi Equivalence). Let θ_G be the set of linearly independent parameters needed to parameterize any distribution $\Theta \in \Theta(G)$. For two directed graphs G_1 and G_2, let μ be the Lebesgue measure defined over $\theta_{G1} \bigcup \theta_{G2}$. G_1 and G_2 are quasi equivalent if $\mu(\theta_{G1} \bigcap \theta_{G2}) \neq 0$.

Assumption 3. (Generic Faithfulness). The probability distribution generated by the underlying model is g-faithful (generically faithful) to the ground truth structure G^*. Specifically, this implies that within the parameter space, the subset of parameter values that result in unfaithful distributions, where conditional independence occurs without d-separation in G^*, has Lebesgue measure zero (Spirtes et al., 2001). This assumption ensures that unfaithful distributions are measure-theoretic anomalies, providing the theoretical consistency required for score-based structural search.

Assumption 4. Let $E(G)$ be the set of edges of G, and let $H(G)$ denote the set of hard constraints on the precision matrix imposed by the graph structure G. For a DAG G^* and a directed graph $\hat{G}$, we have the following statements.
(*a*) If $|E(\hat{G})| \leq |E(G^*)|$, then $H(\hat{G}) \not\subset H(G^*)$.
(*b*) If $|E(\hat{G})| < |E(G^*)|$, then $H(\hat{G}) \not\subseteq H(G^*)$.

Definition 3 defines the concept of equivalence among directed graphs called quasi equivalence (Ghassami et al., 2020). For a directed graph $G, \Theta(G)$ denotes the set of valid precision matrices corresponding to it that parameterize the set of distributions that are Markovian with respect to G's structure. Furthermore, the set of values satisfying hard constraints in this parameter space has Lebesgue measure zero. Definition 3 implies that if directed graphs G_1 and G_2 are quasi-equivalent, then they share the same hard constraints. Assumptions 3 and 4 indicate that in DAGs utilizing edges as parameters will introduce an independent dimension to the distribution space, where $H(\hat{G})$ denotes the set of hard constraints of a directed graph $\hat{G}$ If a DAG has the same number of edges as another directed graph $\hat{G}$, then the distribution space of the DAG cannot be a strict subset of the distribution space of $\hat{G}$ Therefore, when Assumptions 3 and 4 are satisfied, the WIR constraint yields a DAG that is quasi equivalent to the ground truth DAG and has the same number of edges, which is stated in Theorem 2 as follows.

Theorem 2. *If the underlying DAG satisfies Assumptions 3 and 4, then the maximum likelihood estimator with DAG penalty and CI regularization penalty asymptotically outputs a DAG that is quasi equivalent to the ground truth DAG and has the same number of edges.*

The proof of Theorem 2 is presented in Appendix C. Theorem 2 implies that, under mild assumptions, one only has to apply the DAG constraint and the WIR constraint to the likelihood-based objective will return an estimated graph that will be a quasi equivalent to the ground truth DAG.

3.5. Algorithm

Theories 1 and 2 ensure that incorporating our WIC or WIR constraints can converge to a feasible solution and learn a DAG. The overall algorithm of our method proceeds in four main steps: 1) Constructing the CI matrix M from data using low-order CI tests; 2) Formulating the constrained optimization problem (Eq. (7) or Eq. (9)); 3) Solving this problem using an iterative penalty method; 4) Applying a final thresholding step to obtain the sparse DAG. Since the proposed WIC and WIR constraints are derived solely from CI information, our method can be integrated into most existing continuous optimization based methods. Although Eqs. (7) and (9) differ in their objective functions and equality constraints, they are fundamentally Equality-Constrained Programs (ECPs) and can be addressed using general numerical optimization methods. However, since the constraint $W : \mathcal{L}_{DAG}(W) = 0$ is a nonconvex constraint, both Eqs. (7) and (9) are nonconvex programs, thereby inheriting the challenges associated with nonconvex optimization. To solve this, we employ the Quadratic Penalty Method (QPM) (Ng et al., 2022a). QPM transforms the constrained problem into a sequence of unconstrained subproblems. It does this by adding the DAG constraint to the main objective, penalizing $S(W;\phi)$ by a penalty coefficient ρ that increases at each iteration. This approach is essentially the same as the Augmented Lagrangian Method (ALM), but without the Lagrangian part, which has a simpler process. We begin by introducing the quadratic penalty function:

$$\mathcal{Q}(W;\rho,\phi) \triangleq \mathcal{S}(W;\phi) + \frac{\rho}{2}\mathcal{L}_C(W)^2, \tag{10}$$

where $\mathcal{S}(W;\phi)$ denotes the score function, i.e., $\mathcal{L}_{rec}(W;X,\phi) + \lambda_1 \mathcal{L}_{sparse}(W)$, meanwhile, $\mathcal{L}_C(W)$ denotes specific hard constraints for the WIC or WIR methods. For WIC, $\mathcal{L}_C(W)$ is $\mathcal{L}_{DAG}(W) + \lambda_{CI}\|M \circ W\|_F^2$, and for WIR, it is $\mathcal{L}_{DAG}(W)$. Then we apply a numerical optimization algorithm (e.g., Adam [48]) to minimize the penalty function $\mathcal{Q}(W;\rho,\phi)$ as $\rho \to \infty$. By gradually increasing the penalty coefficient ρ, we increase the penalty for constraint violations. Theorem 1 verifies that when we bring ρ tends to infinity, if the DAG constraint term $\mathcal{L}_{DAG}(W)$ satisfies Assumption 2, the process is guaranteed to converge to a feasible solution under mild conditions. When the process terminates and converges to a feasible solution, we obtain the final causal graph. Algorithm 2 describes this QPM procedure. The first stage involves performing CI tests on the observed variables and obtaining the weighted superstructure related to the observations (lines 1–5). In the second phase, we introduce the weighted CI constraints to the continuous optimization problem to solve $\min_W \mathcal{Q}(W;\rho,\phi)$ (line 6). In the third phase, we obtain the minimized $\mathcal{Q}(W;\rho,\phi)$ sequence by gradually increasing ρ (lines 7–14). Finally, we set a threshold of τ to 0.3, rather than strictly requiring $\mathcal{L}_{DAG}(W) = 0$, and connect the edges corresponding to different variables based on the final solution W^* to obtain the causal graph (lines 15–18).

Algorithm 2 QPM for continuous optimization with constraints based on weighted superstructure.

Require: observed data X, the maximum order of CI test k_{CI}, maximum number of iterations $iter_{\max}$, threshold τ, multiplicative factor $\beta > 1$, nonnegative sequence τ_k, starting point ϕ_0, convergence indicator $\epsilon > 0$.
Ensure: causal graph G.
1: **for** $\forall v_i, v_j \in V$ **do**
2:     **if** $v_i \perp\!\!\!\perp v_j \mid Z$ ($\forall Z \subseteq$ one-hop neighbours of $v_i \cup v_j$, $|Z| \le k_{CI}$ and $v_i, v_j \notin Z$) **then**
3:         set $M_{i,j} = M_{j,i} = p$
4:     **else**
5:         set $M_{i,j} = M_{j,i} = 0$
6:     **end if**
7: **end for**
8: Initialize W_0, ρ_0, ϕ_0 and formulate $\mathcal{Q}(W;\rho,\phi)$ with WIC/WIR constraints
9: **for** $k = 1, 2, \ldots, iter_{\max}$ **do**
10:     $W_k = W_{k-1}, \phi_k = \phi_{k-1}$
11:     Solve $\min_{W,\phi} \mathcal{Q}(W;\rho_k,\phi_k)$ to find (W_k,ϕ_k) satisfying $\|\nabla \mathcal{Q}(W_k;\rho_k,\phi_k)\| \le \tau_k$
12:     **if** $\mathcal{L}_{DAG}(W_k) \le \epsilon$ **then**
13:         **return** approximate solution $W^* = W_k$. ▷ Returns weighted adjacency matrix W^* where $W^*_{i,j} \neq W^*_{j,i}$ implies directionality
14:     **end if**
15:     Update penalty coefficient $\rho_{k+1} = \beta \cdot \rho_k$
16: **end for**
17: **for** $\forall v_i, v_j \in V$ **do**
18:     **if** $|W^*_{i,j}| > \tau$ **then**
19:         Connect $v_i \to v_j$ in G.
20:     **end if**
21: **end for**
22: **return** G

3.6. Theoretical analyses

Time complexity: The WIC/WIR procedure, as outlined in Algorithm 2, consists of two main phases: the initial construction of the CI matrix M and the subsequent continuous optimization. First, the complexity of constructing M depends on the number of CI tests. Given d variables and a maximum conditional set size of $k_{\max}$, the number of CI tests used in our method is $O(C_d^2 \cdot \sum_{i=0}^{k_{\max}} C_{d-2}^i)$, which is bounded by $O(\frac{d^{k_{\max}+2}}{(k_{\max}!)})$. Second, regarding the optimization phase, standard methods like NOTEARS or GOLEM are computationally dominated by the acyclicity constraint and its gradient computation (typically involving matrix exponential), which incurs $O(d^3)$ complexity per iteration. In contrast, our proposed WIC/WIR constraints introduce an additional term $\|M \circ W\|_F^2$. Computing this term and its gradient involves only element-wise Hadamard products, resulting in a complexity of $O(d^2)$. Since $O(d^2) \ll O(d^3)$, the additional computational burden imposed by WIC/WIR during the optimization stage is negligible. Consequently, the overall asymptotic complexity per iteration remains consistent with the baseline continuous optimization methods.

Stability and correctness: In the first phase of our algorithm, we calculate the CI matrix M using the CI test. Similar to constraint-based

Table 1
Attributes of the datasets.

Dataset	Nodes	Arcs	Avg degree	Max in degree
Alarm	37	46	2.49	4
Hailfinder	56	66	2.36	4
Win95pts	76	112	2.95	7
Sachs	11	17	3.09	3

causal discovery, the CI test retains or removes edges based on the significance level α. To avoid the irrecoverable errors that can arise from incorrectly removing a true edge, we employ stricter statistical tests during the initial phase of building the CI matrix M. Furthermore, because high-order CI tests will negatively impact both computational efficiency and performance of the CI matrix, we limit the maximum order of CI tests to 1. We then set the significance level α to 0.01 for 0-order tests and 0.1 for 1-order tests to achieve higher precision and stability. The third phase of our algorithm requires solving a continuous optimization problem, which is different from previous hybrid methods such as MMHC (Tsamardinos et al., 2006). MMHC finds a superstructure that strictly limits the subsequent search, which will significantly impair the performance of the optimal optimization result if the superstructure is false. In our method, λ_{CI} is used to balance the influence of WIR constraints on the optimization results, making the method less sensitive to potential errors in the CI matrix. Finally, we set a threshold τ to 0.3 rather than zero, which is more robust to estimation errors. The robustness experiments detailed in Section 4.6 confirm the stability of our method.

4. Performance evaluation

In this section, we evaluate the performance of our proposed method on both simulations and real-world datasets. For simulations, we follow the data-generation process presented in previous studies (Xia et al., 2023) based on three well-known real-world causal network structures $Alarm$, $Hailfinder$, and $Win95pts$ [1]. In real-world scenarios, we employ the Sachs protein signaling network (Sachs et al., 2005), which is widely used for evaluating causal discovery methods. The attributes for these networks are presented in Table 1. Each synthetic data sample x is generated based on the ground truth DAG using either linear or non-linear mechanisms.

Section 4.1 introduces the details of the baselines used for the experiments. In Section 4.2, we compare GOLEM (Ng et al., 2020) with GOLEM + WIR on a generalized linear model. In Section 4.3, we compare DAG-GNN (Yu et al., 2019) with DAG-GNN + WIC/WIR on non-linear model. In Section 4.4, we study the differences between DAG-GNN + WIC/WIR and DAG-GNN + CIC/CIR (Xia et al., 2023). In Section 4.5, we evaluate the proposed method against other classic continuous optimization methods. In Section 4.6, we conduct sensitivity experiments on 1) GOLEM + WIR v.s. GOLEM and GOLEM + CIR, 2) DAG-GNN + WIC v.s. DAG-GNN and DAG-GNN + CIC and 3) DAG-GNN + WIR v.s. DAG-GNN and DAG-GNN + CIR to compare the robustness of different continuous optimization algorithms to handle Type II errors. In Section 4.7, we evaluate the proposed method on a real-world protein dataset $Sachs$ (Sachs et al., 2005). Across all experiments, the causal discovery performance is assessed using two metrics: the F1 score (the harmonic mean of recall and precision) and structural Hamming distance (SHD, the total number of missing, falsely detected, or reversed edges).

4.1. Baseline details

The implementation details of the continuous optimization baselines are listed below:

- DAG-GNN (Yu et al., 2019). We use the code available at the GitHub repository https://github.com/fishmoon1234/DAG-GNN. In the proposed WIC/WIR constrained version, we use QPM instead of ALM to ensure convergence, setting the Lagrangian multiplier $\lambda = 0$. The encoder and decoder sizes are set to 96.
- DAG-GNN + CIC/CIR (Xia et al., 2023). We separately add a CI hard constraint CIC and a regularization term CIR on DAG-GNN. The code is sourced from the GitHub repository available at https://github.com/xyw5vplus1/CIR. The CI coefficient is fixed at 1.0.
- GOLEM (Ng et al., 2020). We use the code available at the GitHub repository https://github.com/ignavierng/golem. We follow the initialization scheme that optimizes GOLEM-NV objective using the solution returned by GOLEM-EV objective. The parameters are set to $\lambda_1 = 0.05, \lambda_{DAG} = 5.0, \lambda_{CI} = 0.5$ in GOLEM-NV and the parameters are set to $\lambda_1 = 0.1, \lambda_{DAG} = 5.0, \lambda_{CI} = 1.0$ in GOLEM-EV.
- GOLEM + CIR (Xia et al., 2023). As for CIR, we use the code available at the GitHub repository https://github.com/xyw5vplus1/CIR. The CI coefficient is set to 1.0.
- NOTEARS (Zheng et al., 2018). We use the code from the GitHub repository https://github.com/xunzheng/notears/blob/master/notears/linear.py. The threshold is set to 0.3 as default.
- NOTEARS-MLP (Zheng et al., 2020). We use the code from the GitHub repository https://github.com/xunzheng/notears/blob/master/notears/nonlinear.py. The threshold is set to 0.3 as default.
- NOTEARS-SOB (Zheng et al., 2020). We use the code from the GitHub repository https://github.com/huawei-noah/trustworthyAI/blob/master/gcastle/castle/algorithms/gradient/notears/torch/nonlinear.py. The threshold is set to 0.3 as default.
- MCSL-MLP (Ng et al., 2022b). We use the code from the GitHub repository https://github.com/huawei-noah/trustworthyAI/blob/master/gcastle/castle/algorithms/gradient/mcsl/torch/mcsl.py. The threshold is set to 0.7 as default.
- GAE (Ng et al., 2019). We use the code from the GitHub repository https://github.com/huawei-noah/trustworthyAI/blob/master/gcastle/castle/algorithms/gradient/gae/torch/gae.py. The threshold is set to 0.3 as default.
- GraN-DAG (Lachapelle et al., 2020). We use the code from the GitHub repository https://github.com/huawei-noah/trustworthyAI/blob/master/gcastle/castle/algorithms/gradient/gran_dag/torch/gran_dag.py.

4.2. Performance on linear cases

We firstly examine the extent to which the WIR constraint can enhance the performance of GOLEM (Ng et al., 2020) in linear scenarios. The synthetic data used in the experiment is generated according to the linear SEM: $x = W^T x + z$. The edge weights in W are sampled uniformly from $(-2.0, -0.5) \cup (0.5, 2.0)$. Considering the heterogeneity of noise, the exogenous noise z_i is sampled from $\mathcal{U}(-1.0, 1.0)$ for root nodes and $\mathcal{U}(-0.2, 0.2)$ for other nodes. We employed partial correlation with Fisher's z-transform as our CI test method. The two main factors that influence the weighted superstructure are sample size and the maximum order of CI tests. We first set the maximum order of CI tests to 1 and then adjust the significance level for different orders. For the 0-order CI test, we set the significance level α to 0.01. For the 1-order CI test, we set the significance level α to 0.1. Finally, we set the sample size to 200, 500 and 1000. The error threshold τ of the continuous optimization algorithm is set to 0.3.

Given that GOLEM is a fully soft constrained method, we only add the WIR constraint to it. To alleviate the local optimality problem, we first initialize using the equal noise variance (likelihood-NV) objective function and then fine-tune the solution using the unequal noise variance (likelihood-EV) objective function. For simplicity, we denote GOLEM with k-order WIR constraints as GOLEM + WIR(k).

The experimental results are illustrated in Fig. 1, demonstrating that our WIR constraint consistently improves the causal discovery

[1] Causal graphs for the $Alarm$, $Hailfinder$ and $Win95pts$ are available at the bnlearn repository: https://www.bnlearn.com/bnrepository/

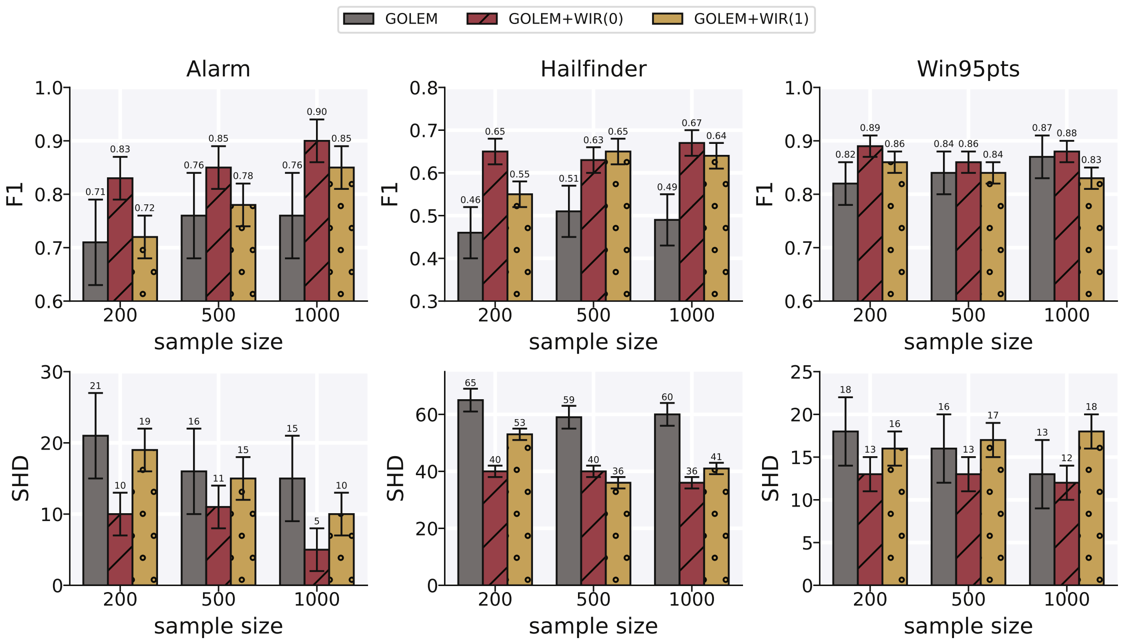

Fig. 1. Experiment results on linear cases. The performance of GOLEM and GOLEM with 0-order and 1-order WIR constraints is evaluated using F1 score and SHD across sample sizes of $\{200, 500, 1000\}$.

performance of GOLEM across different sample sizes in terms of F1 score and SHD, showcasing the effectiveness of WIR in handling heterogeneous noise. Notably, the 0-order WIR significantly enhances performance with smaller sample sizes. Specifically, compared to the baseline GOLEM, the SHD of GOLEM + WIR(0) is reduced by 52.3% on $Alarm$, 38.46% on $Hailfinder$, and 27.78% on $Win95pts$. This improvement is attributed to the 0-order CI test, which can more effectively reduce the Type II error when the sample size is small.

Furthermore, in most cases, GOLEM + WIR(0) outperforms GOLEM + WIR(1), indicating that simply increasing the order of the CI tests does not guarantee better performance. This is because the 1-order CI test on limited samples may introduce Type II errors, leading to the incorrect penalization of the true edges. This effect was particularly pronounced in the $Win95pts$ network with 1000 samples, where we observed that the performance of GOLEM + WIR(1) is slightly lower than the GOLEM method. This anomalous result can be attributed to the unique structural properties of the $Win95pts$ network itself. Compared to $Alarm$ and $Hailfinder, Win95pts$ is significantly larger and denser (nodes are 37 and 56 vs. 76; average degree is 2.49 and 2.36 vs. 2.95). In such cases, high-order CI tests may inevitably remove some true edges, thereby increasing the SHD. Therefore, this result demonstrates the risks of applying 1-order CI tests to complex graphs and highlights the robustness of the WIR(0) approach, which relies only on the most statistically reliable 0-order tests.

4.3. Performance on non-linear cases

In this section, we evaluate the causal discovery performance of the proposed WIC and WIR constraints in conjunction with DAG-GNN (Yu et al., 2019) in nonlinear scenarios. The synthetic data used in the experiment is generated using the DAG-GNN method: $x = 2sin(W^T(x + 0.5 \cdot \mathbf{1})) + W^T(x + 0.5 \cdot \mathbf{1}) + z$. Where $\mathbf{1}$ is a vector with all elements equal to 1. This generative method considers more challenging and complex nonlinear generative models. Generally, the nonlinearity occurs before the linear combination of the variables, whereas this method applies nonlinearity after the linear combination of the variables. All other experimental settings, including the exogenous noise z_i, CI test significance levels ($\alpha = 0.01$ for 0-order, $\alpha = 0.1$ for 1-order), and sample sizes (200, 500, 1000), are consistent with the linear experiments. For convenience, we use DAG-GNN + WIC/WIR(k) to represent DAG-GNN with k-order WIC and WIR constraints.

The experimental results are shown in Fig. 2. Compared with the baseline DAG-GNN, the proposed method WIC/WIR can significantly improve the causal discovery performance in most cases. When the sample size is 200 and the 0-order CI test is used, DAG-GNN + WIC(0)/WIR(0) reduces the SHD by 38.42%/47.37%, 49.39%/47.56% and 65.58%/69.54% on *Alarm,Hailfinder* and *Win95pts*, respectively, compared to DAG-GNN. When the 1-order CI test is used, DAG-GNN + WIC(1)/WIR(1) reduces the SHD by 58.95%/51.58%, 58.54%/52.44%, and 66.23%/72.08%, respectively, demonstrating the effectiveness of WIC/WIR. In contrast to linear scenarios, in non-linear scenarios, DAG-GNN + WIC/WIR(1) generally outperforms DAG-GNN + WIC/WIR(0) in most cases. This is because DAG-GNN trains neural networks to capture complex nonlinear structures of data, and the 1-order CI test can effectively provide the crucial evidence needed to exclude the specific non-causal edges and improve its performance in nonlinear situations.

Additionally, we also observe that DAG-GNN + WIC is not always outperforming DAG-GNN + WIR on different datasets. For example, on the *Hailfinder* network, when the sample size is 200, WIR performs better than WIC. When the sample size increases, the performance of WIR and WIC is close. This is because, with a smaller sample size, the CI test is more prone to Type II errors, which can cause the WIC constraint to incorrectly eliminate true edges. In contrast, the regularization coefficient λ_{CI} of the WIR constraint is still capable of keeping these edges when the sample size is limited, thus reducing the negative impact of CI. As the sample size increases, the Type II error rate of the CI test decreases, and the performance of WIC and WIR converges, becoming nearly identical.

4.4. Performance comparison with the CIC/CIR method

In this section, we evaluate the causal discovery performance of DAG-GNN + CIC/CIR and DAG-GNN + WIC/WIR on the $Alarm$ network. The synthetic data for the experiment are generated using the DAG-GNN method: $x = 2sin(W^T(x + 0.5 \cdot \mathbf{1})) + W^T(x + 0.5 \cdot \mathbf{1}) + z$. We use the same hyperparameters for the comparison method. For the 0-order CI test, we set the significance level α to 0.01. For the 1-order CI test, we

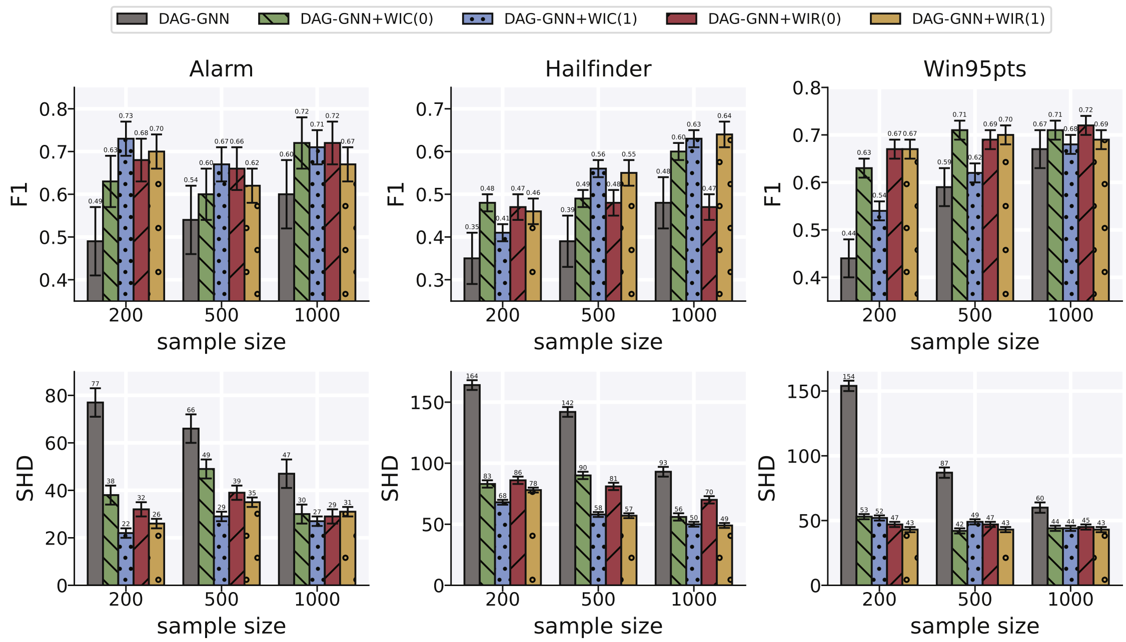

Fig. 2. Experiment results on non-linear cases. The performance of DAG-GNN and DAG-GNN with 0-order and 1-order WIC/WIR is evaluated using F1 score and SHD across sample sizes of $\{200, 500, 1000\}$.

set the significance level α to 0.1. The sample size is set to 200 and 1000. For convenience, we use DAG-GNN + CIC/CIR/WIC/WIR(k) to represent DAG-GNN with k-order CIC, CIR, WIC and WIR constraints. Performance is evaluated using four metrics: precision, recall, F1 score, and SHD.

The experimental results are shown in Fig. 3. Compared to CIC/CIR, the proposed WIC/WIR method significantly enhances the performance of Precision, F1 score and SHD at both sample sizes. Specifically, for SHD, the 0-order WIC/WIR reduces by 22.72%/16.36%, and the 1-order WIC/WIR reduces by 29.89%/10.94%. With a sample size of 200, WIC outperforms CIC in terms of SHD and F1. This indicates that, while the CIC constraint can identify many potential edges, the directions of these edges are not necessarily accurate. In contrast, WIC predicts fewer edges overall and achieves higher accuracy in predicting edge directions, as reflected in the Precision metric.

Furthermore, the WIC/WIR has a slightly lower Recall than the CIC/CIR methods. This is because WIC/WIR utilizes a weighted superstructure, which imposes stricter edge evaluation during the optimization process, leading to a slightly lower Recall by sacrificing a small number of true edges in exchange for being more effective at filtering out false positives. The performance of DAG-GNN + WIC/WIR(k) also depends on the sparsity of W determined by M. As argued in Zhang et al. (2011) and verified by our experiments, low-order CI tests (controlling set $|Z| \in \{0, 1\}$) offer the best balance of reliability and computational efficiency. Therefore, we believe that restricting our method to these robust, low-order tests is sufficient to support robust continuous optimization based causal discovery.

4.5. Performance comparison with other continuous optimization methods

In this section, we evaluate our method by incorporating WIC/WIR constraints in conjunction with six continuous optimization methods, including NOTEARS (Zheng et al., 2018), NOTEARS-MLP (Zheng et al., 2020), NOTEARS-SOB (Zheng et al., 2020), MCSL-MLP (Ng et al., 2022b), GAE (Ng et al., 2019), and GraN-DAG (Lachapelle et al., 2020). We evaluate performance on the $Alarm$ network across sample sizes of 200, 500 and 1000. The results are shown in Fig. 4. We can see that under different sample sizes, all methods show the improvement of SHD and F1 by incorporating WIC/WIR constraints. However, the performance improvement of different methods after adding WIC or WIR constraints varies slightly. For NOTEARS-SOB, with a sample size of 200, the incorporation of WIC constraints significantly improves the SHD with only a little gain in F1. This indicates that WIC effectively eliminates many false edges while sacrificing a small portion of the true edges. For MCSL-MLP and NOTEARS with the same sample size of 200, although their performance is already commendable, the incorporation of WIC can still significantly improve their performance. When the sample size is 200, GAE performs better without WIC constraints. However, as the sample size increases to 500 and 1000, the method with WIC constraints significantly outperforms the original method in terms of SHD. This demonstrates that CI tests may mislead a powerful model like GAE on limited data. However, when the sample size is large enough to provide reliable CI information, WIC can effectively improve the accuracy of GAE. Furthermore, the highly complex, regularized models like MCSL-MLP and GAE show no significant improvement after adding the WIR constraint. This is because the WIR constraint, as an additional mild soft constraint, tends to regularize the model and may not have a sufficient impact on the model. In contrast, the WIC will hard constraint on the model and can still effectively improve the performance of these methods.

4.6. Robustness on type II error of CI

Here, we evaluate the extent to which WIC and WIR can improve the robustness of different continuous optimization algorithms in both linear and nonlinear scenarios. We introduce random errors by randomly altering the values of some elements in the CI matrix M after generating it in the first phase of the algorithm. Specifically, we change the value $M_{i,j}$ in the CI matrix M from 0 to 1, while the ground truth is $M_{i,j} = 0$. We employ only 0-order CI tests in both scenarios for two primary reasons. First, the number of 0-order CI tests remains constant, whereas the number of 1-order tests is not fixed, making comparisons less controlled. Second, the robustness experiment aims to compare the stability of the algorithm when errors occur in the CI test. Using 1-order tests makes it impossible to clearly attribute the removal of edges to specific injected errors in either 0-order or 1-order CI tests.

We first conducted sensitive experiments with GOLEM on the $Alarm$ network in linear scenarios. For simplicity, GOLEM + WIR(0) + t error(s) indicates that there are t errors introduced to M, where

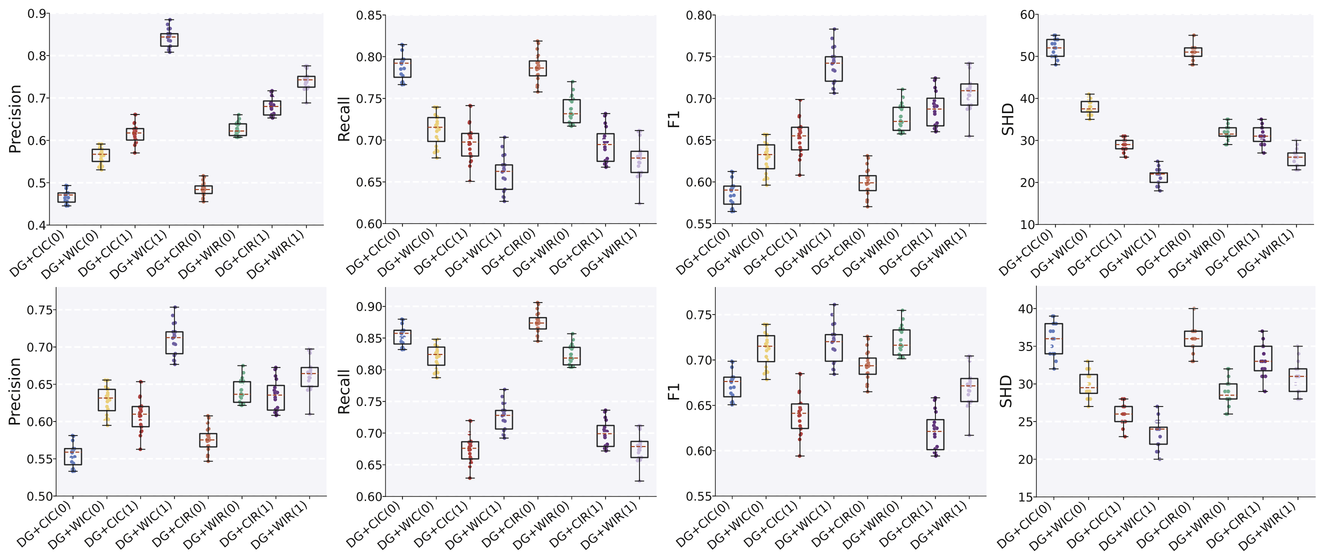

Fig. 3. Experiment results on the *Alarm* network. The performance of DG (DAG-GNN) with 0-order and 1-order CIC/CIR/WIC/WIR constraints is displayed across four metrics-Precision, Recall, F1 score, and SHD from left to right-for sample sizes of 200 (top) and 1000 (bottom).

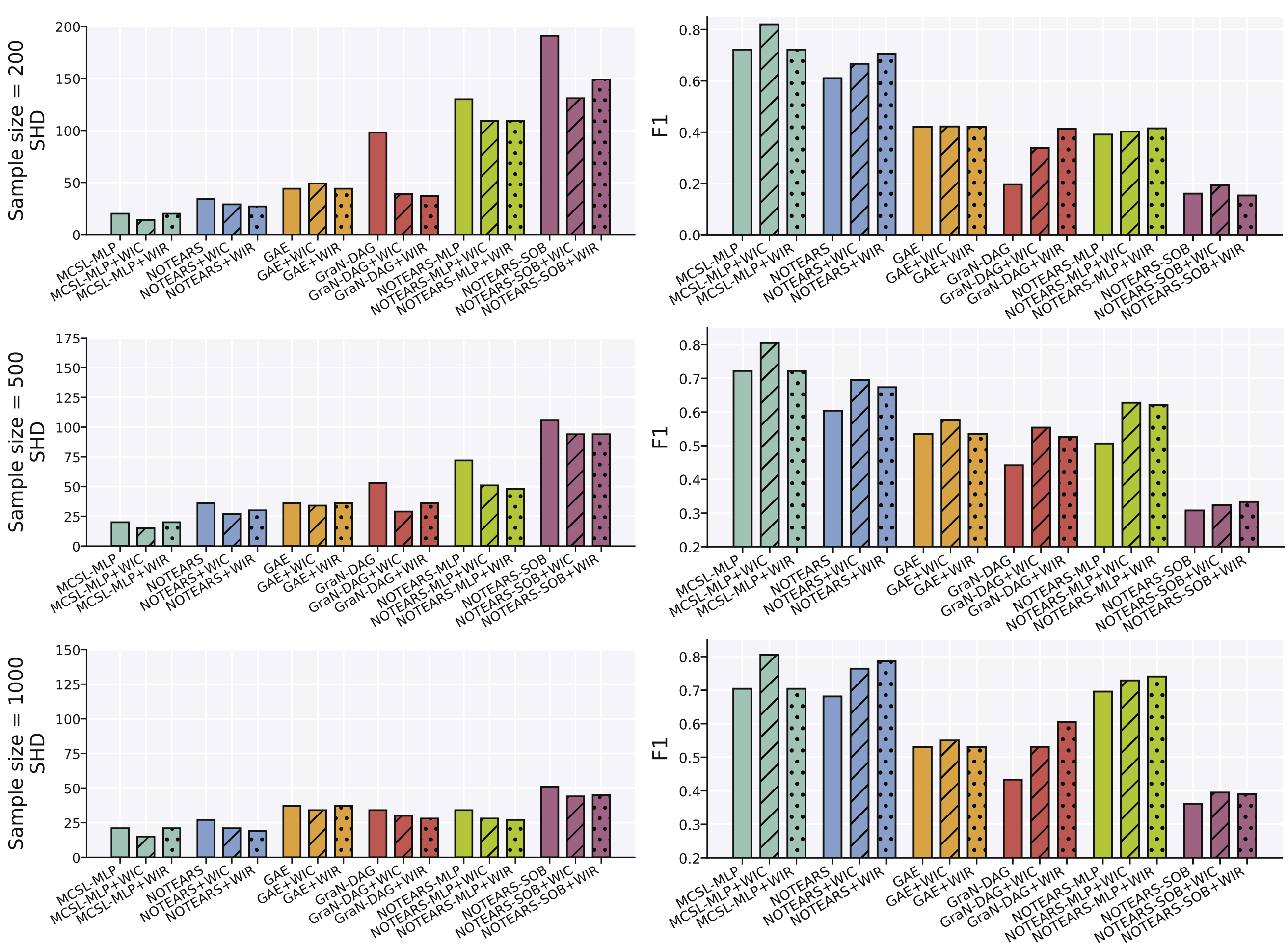

Fig. 4. Causal discovery results of different continuous optimization methods incorporating WIC/WIR constraints on $Alarm$. The sample size is set to 200 (Top), 500 (Middle) and 1000 (Bottom).

Table 2
Robustness results of WIR against type II error rates in CI tests for linear cases with 1000 samples on alarm network.

Method	F1 score	SHD
GOLEM	0.68 ± 0.08	22.4 ± 6.18
GOLEM + CIR(0) + 0 error	$\mathbf{0.85 \pm 0.04}$	10.6 ± 3.38
GOLEM + WIR(0) + 0 error	$\mathbf{0.85 \pm 0.04}$	$\mathbf{9.6 \pm 3.21}$
GOLEM + CIR(0) + 1 error	0.79 ± 0.04	15.4 ± 4.32
GOLEM + WIR(0) + 1 error	0.76 ± 0.05	15.2 ± 4.22
GOLEM + CIR(0) + 2 errors	0.75 ± 0.07	17.4 ± 4.84
GOLEM + WIR(0) + 2 errors	0.73 ± 0.06	17.2 ± 4.67
GOLEM + CIR(0) + 3 errors	0.74 ± 0.06	19.0 ± 5.69
GOLEM + WIR(0) + 3 errors	0.70 ± 0.06	21.4 ± 5.88
GOLEM + CIR(0) + 4 errors	0.69 ± 0.07	22.4 ± 6.22
GOLEM + WIR(0) + 4 errors	0.68 ± 0.07	22.6 ± 6.34
GOLEM + CIR(0) + 5 errors	0.67 ± 0.07	25.0 ± 7.04
GOLEM + WIR(0) + 5 errors	0.61 ± 0.07	27.6 ± 7.44

Table 3
Robustness results of WIC against type II error rates in CI tests for non-linear cases with 1000 samples on alarm network.

Method	F1 score	SHD
DAG-GNN	0.60 ± 0.08	46.7 ± 6.18
DAG-GNN + CIC(0) + 0 error	0.63 ± 0.07	41.0 ± 4.65
DAG-GNN + WIC(0) + 0 error	$\mathbf{0.64 \pm 0.06}$	$\mathbf{39.9 \pm 4.22}$
DAG-GNN + CIC(0) + 1 error	0.60 ± 0.04	45.0 ± 4.32
DAG-GNN + WIC(0) + 1 error	0.62 ± 0.04	42.0 ± 4.42
DAG-GNN + CIC(0) + 2 errors	0.57 ± 0.07	48.8 ± 4.84
DAG-GNN + WIC(0) + 2 errors	0.59 ± 0.05	44.9 ± 4.65
DAG-GNN + CIC(0) + 3 errors	0.58 ± 0.06	48.0 ± 5.69
DAG-GNN + WIC(0) + 3 errors	0.57 ± 0.06	47.0 ± 4.89

Table 4
Robustness results of WIR against type II error rates in CI tests for non-linear cases with 1000 samples on alarm network.

Method	F1 score	SHD
DAG-GNN	0.60 ± 0.08	46.7 ± 6.18
DAG-GNN + CIR(0) + 0 error	0.63 ± 0.07	40.8 ± 4.45
DAG-GNN + WIR(0) + 0 error	$\mathbf{0.66 \pm 0.05}$	$\mathbf{37.6 \pm 3.78}$
DAG-GNN + CIR(0) + 1 error	0.61 ± 0.05	44.3 ± 4.82
DAG-GNN + WIR(0) + 1 error	0.64 ± 0.06	39.3 ± 4.24
DAG-GNN + CIR(0) + 2 errors	0.58 ± 0.05	47.8 ± 3.54
DAG-GNN + WIR(0) + 2 errors	0.61 ± 0.05	42.6 ± 4.55
DAG-GNN + CIR(0) + 3 errors	0.58 ± 0.04	48.3 ± 3.96
DAG-GNN + WIR(0) + 3 errors	0.58 ± 0.04	46.8 ± 4.43

Table 5
Experiment results on sachs dateset.

Method	#Total	#Correct	SHD
PC	14	8	15
ICA-LiNGAM	8	4	14
MMHC	8	4	13
CAM	10	6	12
NOTEARS	20	6	19
GOLEM	11	6	14
RL-BIC	10	7	**11**
DARING	15	7	**11**
GraN-DAG	10	5	13
NOTEARS-MLP	11	6	**11**
DAG-GNN	15	6	16
DAG-GNN + CIC/CIR	9	6	**11**
DAG-GNN + WIC/WIR	9	6	**11**

$t = 1, 2, \ldots, 5$. We fixed the sample size as 1000 and set the significance level $\alpha = 0.01$. The experimental results are listed in Table 2. The results of 10 simulations show that our method outperforms the original GOLEM when t is less than 5, which verifies the robustness of our method. Note that low-order CI tests can usually provide reliable performance with fewer samples, which allows our method can improve the performance of existing continuous optimization methods in small sample size scenarios. In addition, the average number of incorrectly removed edges in the CI test for our method is 1.06, compared to 1.8 for GOLEM + CIR, indicating that our method is more robust than the CIR method.

We then conducted similar experiments with DAG-GNN on the *Alarm* network in nonlinear scenarios. For simplicity, we denote DAG-GNN + WIC/WIR(0) + t error(s) as introducing t errors into M, where $t = 1, 2, \ldots, 3$. We fixed the sample size as 1000 and set the significance level $\alpha = 0.01$. The experimental results are presented in Tables 3 and 4. The results of 10 simulations demonstrate that our method outperforms the original DAG-GNN when t is less than 3. The average number of incorrect edge cuts for DAG-GNN + CIC/CIR is 3.33, whereas the average number of incorrect edge cuts for the proposed method is 3.2. Combined with the results of the F1 score and SHD, this demonstrates that the proposed method is more robust than the CIC/CIR method. Moreover, the F1 score of WIR is higher than that of WIC, and the SHD is lower. This can be attributed to the regularization coefficient, which provides more robustness to the estimation error of the WIR constraint, which is consistent with our statement in Section 3.6.

4.7. Performance on real-world data

In this section, we evaluate the causal discovery performance of the proposed method on the Sachs dataset (Sachs et al., 2005), a well-known dataset related to real-world protein signaling networks. This dataset is commonly used in the literature on probabilistic graphical models, with experimental annotations validated by the biological community. Since this dataset includes ground truth, it has been widely used in previous studies (He et al., 2021). This dataset comprises 853 samples and a corresponding causal graph [2] with 11 nodes and 17 arcs. These nodes form 55 possible pairs, of which 17 pairs are independent and the remaining 38 are not. In this experiment, we apply WIC/WIR to DAG-GNN (Yu et al., 2019). We use 12 existing methods as baselines, including 8 continuous optimization-related methods: NOTEARS (Zheng et al., 2018), NOTEARS-MLP (Zheng et al., 2020), GraN-DAG (Lachapelle et al., 2020), RL-BIC (Zhu et al., 2020), GOLEM (Ng et al., 2020), DARING (He et al., 2021), DAG-GNN (Yu et al., 2019), and DAG-GNN + CIC/CIR (Xia et al., 2023); 2 classic methods: PC (Spirtes et al., 2001) and ICA-LiNGAM (Shimizu et al., 2006); and 2 super-structure-based methods: MMHC (Tsamardinos et al., 2006) and CAM (Ng et al., 2021). It is important to note that the authors of DARING did not release their code, so we directly used the results from the original paper for the Sachs dataset.

The experimental results are shown in Table 5. Among all the methods compared, DAG-GNN + WIC/WIR achieved the lowest SHD of 11, RL-BIC, DARING, DAG-GNN + CIC/CIR and NOTEARS-MLP also achieved the same results. Compared with DAG-GNN, the inclusion of WIC and WIR constraints reduced the number of false edges by 5, significantly improving the prediction accuracy and reducing the SHD. We further present our estimated graph in Fig. 5. It can be observed that the graph is acyclic and contains no undirected edges. Our method learns a total of 9 edges, successfully identifying 6 out of 17 ground truth edges (marked by blue arrows), predicting 3 reverse edges (marked by green arrows), and failing to predict the remaining 8 edges (marked by black arrows). Note that DAG-GNN + WIC and DAG-GNN + WIR yield same SHD scores on this dataset. This is attributed to the small scale of the Sachs network, where both hard constraints and soft regularization effectively guide the model to the same sparse structure. These results show the effectiveness and superiority of WIC and WIR on real-world data.

[2] Sachs causal graph: https://www.bnlearn.com/bnrepository/.

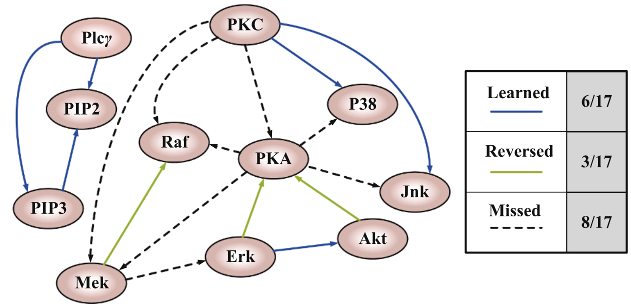

Fig. 5. Results on sachs dataset using WIC/WIR constraint.

5. Discussion

Most existing continuous optimization frameworks for causal discovery predominantly leverage sparsity and functional acyclicity constraints to identify DAGs. Nevertheless, in the presence of limited sample sizes or complex nonlinear dependencies, these rigid constraints often struggle to discern subtle causal signals. This limitation frequently leads to over-pruning of real edges, thereby increasing the Type II error rate-a common defect across current continuous optimization paradigms. To verify the influence of WIC/WIR constraints on continuous optimization algorithms, we conducted many experiments on the combination of GOLEM, DAG-GNN, NOTEARS, NOTEARS-MLP, NOTEARS-SOB, MCSL-MLP, GAE, and GraN-DAG and found that the performance of these baseline methods is inconsistent and highly influenced by Type II errors. Specifically, if prior knowledge of the underlying causal structure of the observations is available, WIC/WIR can generally enhance the robustness of continuous optimization methods to Type II errors and improve their overall performance. Our empirical results further highlight a key trade-off between statistical stability and regularization strength: linear models favor the robustness of 0-order tests against estimation errors, whereas over-parameterized nonlinear models benefit more from the stronger sparsity bias of 1-order tests to constrain their massive search space. Overall, how to effectively reduce Type II errors remains a challenging problem for WIC/WIR and other continuous optimization methods. Furthermore, our weighted framework offers a distinct advantage over traditional "hard" constraints by supporting probabilistic domain knowledge. Instead of absolute certainties, real-world expert priors are often "soft" (e.g., an 80% interaction probability). The matrix M naturally incorporates such priors by mapping high expert confidence to lower penalty weights, thereby selectively relaxing regularization. This positions the weighted superstructure as a flexible weight to balance statistical evidence with expert knowledge. In future work, we plan to extend this approach to more complex settings involving latent confounders and interventional data, which is a crucial direction as highlighted by recent advancements in nonparametric causal estimation with latent variables.

6. Conclusion

Existing causal discovery methods based on continuous optimization are prone to poor performance when encountering heterogeneous noise and small sample sizes, which results in poor accuracy. To address these challenges, we propose a novel continuous optimization method that leverages weighted CI information to improve the performance of causal discovery. Specifically, we construct a weighted superstructure related to the observations using a low-order CI test and derive weighted CI constraints based on this superstructure, which have hard and soft versions. The proposed method is evaluated through extensive experiments with existing continuous optimization algorithms on both synthetic and real-world datasets. Experimental results show that our method can effectively improve the performance of the existing continuous optimization methods, especially in cases of fewer samples.

CRediT authorship contribution statement

Mingjie Chen: Writing - review & editing, Writing - original draft, Software, Methodology, Formal analysis, Conceptualization; **Yewei Xia:** Writing - review & editing, Writing - original draft, Methodology; **Hao Zhang:** Supervision, Funding acquisition; **Ruxin Wang:** Writing - review & editing; **Yuzhong Peng:** Writing - review & editing; **Jihong Guan:** Writing - review & editing; **Shuigeng Zhou:** Supervision, Funding acquisition.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used **Gemini** in order to improve language and readability. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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This *Appendix* is organized as follows:

- Appendix A: Proof of Lemma 2
- Appendix B: Proof of Theorem 1
- Appendix C: Proof of Theorem 2

Appendix A. Proof of Lemma 2

Lemma 3. *Any function $\mathcal{L}_{DAG}(W)$ defined in Eq. (8) satisfies Assumption 2.*

Proof. We first provide acyclicity characterize function $\mathcal{L}_{DAG}(W)$ and its gradient is given by

$$\nabla \mathcal{L}_{DAG}(W) = \sum_{p=1}^{d} C_p \cdot p \cdot (W \circ W)^{p-1^T} \circ 2W. \tag{A.1}$$

If part: We first prove that $\nabla\mathcal{L}_{DAG}(W)=0\rightarrow\mathcal{L}_{DAG}(W)=0$. Consider the diagonal entries W_{ii} for $i\in d$ of $\nabla\mathcal{L}_{DAG}(W)$,

$$\begin{aligned}\nabla\mathcal{L}_{DAG}(W)_{ii}&=\sum_{p=1}^{d}C_p\cdot p\cdot((W\circ W)^{p-1^T})_{ii}\circ 2W_{ii}\\&=(\sum_{p=1}^{d}C_p\cdot p\cdot((W\circ W)^{p-1^T})_{ii})\circ 2W_{ii}.\end{aligned}$$

Since $p\geq 1, c_p>0$ and $W\circ W$ is a non-negative matrix, $\nabla\mathcal{L}_{DAG}(W)_{ii}=0$ implies that $W_{ii}=0$, in other words, $G(W)$ does not have any self-loop. Now we consider the elements that (i,j)th entry of W with $i\neq j$, following the same derivation process we get

$$\nabla\mathcal{L}_{DAG}(W)_{ij}=\sum_{p=1}^{d}C_p\cdot p\cdot((W\circ W)^{p-1^T})_{ji}\circ 2W_{ij}.$$

$\nabla\mathcal{L}_{DAG}(W)_{ij}=0$ indicates that at least one of W_{ij} and $((W\circ W)^{p-1^T})_{ji}$ is zero. If $W_{ij}\neq 0$, then $((W\circ W)^{p-1^T})_{ji}=0,\forall p=1,\dots,d$. If there is a cycle both are non-zero. This contradicts the statement that at least one of W_{ij} and $((W\circ W)^{p-1^T})_{ji}$ is zero, because they have edge weight values in one direction and adjacency matrix values in the other. Therefore, the edge from node i to node j, if exists, must not be part of any cycle. If $W_{ij}=0$, it is trivial to obtain the same conclusion. Since the above conclusion holds for all different i,j. Combining the above cases, we conclude that all edges must not be part of any self-loop or cycle, and that W represents a DAG, i.e., $L_{DAG}(W)=0$.

Only if part: Then we prove that $\mathcal{L}_{DAG}(W)=0\rightarrow\nabla\mathcal{L}_{DAG}(W)=0$. Notice that $\mathcal{L}_{DAG}(W)=0$ implies that W represents a DAG. Since a DAG does not have any self-loop, the diagonal entries of W must be zero, and so are the diagonal entries of $\nabla\mathcal{L}_{DAG}(W)$. Now, we need to consider the (i,j)th entry of $\mathcal{L}_{DAG}(W)$ with $i\neq j$:

- If there is no directed edge from node i to node j, i.e. $W_{ij}=0$, then it is clear that $\sum_{p=1}^{d}C_p\cdot p\cdot((W\circ W)^{p-1^T})_{ji}\cdot 2W_{ij}=0$.
- If there is an edge from node i to node j in a graph, represented by a non-zero weight W_{ij}, i.e. $W_{ij}\neq 0$, then the corresponding entry in other term $((W\circ W)^{p-1^T})_{ji}$ must be zero. Otherwise, a self-loop will be formed, which contradicts the statement that W represents a DAG. Conversely, if $((W\circ W)^{p-1^T})_{ji}$ is non-zero, it implies that there must have a direct edge from node j to node i, hence W_{ij} must be zero. Therefore, at least one of W_{ij} and $((W\circ W)^{p-1^T})_{ji}$ must be zero. Combining the above cases, we have $\nabla\mathcal{L}_{DAG}(W)=0$.

We then complete this proof. □

Appendix B. Proof of Theorem 1

Theorem 1. *Consider the WIC constrained optimization problem in Eq. (7), solved via the QPM. Let the DAG constraint be any function from the class defined in Eq. (8) and the sequence of iterates be $\{W^k\}$. If the penalty coefficients diverge to infinity ($\rho_k\rightarrow\infty$), the non-negative tolerance sequence $\{\tau_k\}$ is bounded, and Assumption 2 holds for the function $\mathcal{L}_{DAG}(W)$, then any limit point W^* of the sequence $\{W^k\}$ will be feasible, corresponding to a feasible DAG structure.*

Proof. Since $\mathcal{L}_{DAG}(W)$ is strictly non-negative, and $M_{i,j}$ and $W_{i,j}$ in the WIC constraint are also strictly non-negative. Therefore, we can combine the DAG constraint with the WIC constraint and transform the whole constraint into $\mathcal{L}_{DAG}(W)+\lambda_{CI}\|M\circ W\|_F^2=0$, where $\circ$ is the matrix Hadamard product, $\|\cdot\|_F$ is the Frobenius norm, and λ_{CI} is a hyper-parameter to balance the influence of WIC constraint term. Notice that in Lemma 2, we have proved that $\mathcal{L}_{DAG}(W)$ can converge to a feasible solution by satisfying Assumption 2. Therefore, for the convergence guarantee of the total constraint defined in Eq. (7), we just need to prove that the total constraint $\mathcal{L}_C(W)=\mathcal{L}_{DAG}(W)+\lambda_{CI}\|M\circ W\|_F^2$ also satisfies Assumption 2. Then, using Lemma 1, we get the convergence guarantee of the total constraint defined in Eq. (7). The total constraint $\mathcal{L}_C(W)$ gradient is given by

$$\begin{aligned}\nabla\mathcal{L}_C(W)=&\sum_{p=1}^{d}C_p\cdot p\cdot(W\circ W)^{p-1^T}\circ 2W\\&+2(M\circ W)\circ W.\end{aligned}\tag{B.1}$$

If part: As in the previous prove process, we first prove $\nabla\mathcal{L}_C(W)=0\rightarrow\mathcal{L}_C(W)=0$. Consider the diagonal items W_{ii} for $i\in d$ of $\nabla\mathcal{L}_C(W)$,

$$\begin{aligned}\nabla\mathcal{L}_C(W)_{ii}=&\sum_{p=1}^{d}C_p\cdot p\cdot((W\circ W)^{p-1^T})_{ii}\cdot 2W_{ii}\\&+2(M_{ii}\circ W_{ii})\circ M_{ii}\\=&2*(\sum_{p=1}^{d}C_p\cdot p\cdot((W\circ W)^{p-1^T})_{ii}+M_{ii}^2)\circ W_{ii}.\end{aligned}$$

Since $M_{ii}^2\geq 0,\nabla\mathcal{L}_C(W)_{ii}=0$ implies that $\forall i=1,\dots,d,W_{ii}=0$, then it is obvious that $G(W)$ does not have any self-loop and is also satisfied $M_{i,i}\circ W_{i,i}=0$. Now we consider the elements that (i,j)th entry of W with $i\neq j$, following the same derivation process we get

$$\nabla\mathcal{L}_C(W)_{ij}=2*(\sum_{p=1}^{d}C_p\cdot p\cdot((W\circ W)^{p-1^T})_{ji}+M_{ij}^2)\circ W_{ij}.$$

$\nabla\mathcal{L}_C(W)_{ij}=0$ indicates that at least one of W_{ij} and $((W\circ W)^{p-1^T})_{ji}+M_{ij}^2$ is zero. If $W_ij\neq 0$, then $((W\circ W)^{p-1^T})_{ji}+M_{ij}^2$ must be zero, $\forall p=1,\dots,d$. If there is a cycle both are non-zero. This contradicts the statement that at least one of W_{ij} and $((W\circ W)^{p-1^T})_{ji}+M_{ij}^2$ is zero, because they have edge weight values in one direction and adjacency matrix values in the other one. Therefore, the edge from node i to node j, if exists, must not be part of any cycle. In addition, since $((W\circ W)^{p-1^T})_{ji}+M_ij^2$ must be zero, therefore $((W\circ W)^{p-1^T})_{ji}$ and M_{ij}^2 are both zero, thus satisfying $M_{ij}\circ W_{ij}=0$. If $M_{ij}=0$, it is trivial to obtain the same conclusion. Since the above conclusion holds for all different i,j. Combining the above cases, we conclude that all edges must not be part of any self-loop or cycle, and that W represents a DAG. The acyclic of W ensure always satisfies $M\circ W=0$, therefore $\mathcal{L}_C(W)=0$.

Only if part: Then we prove that $\mathcal{L}_C(W)=0\rightarrow\nabla\mathcal{L}_C(W)=0$. Notice that $\mathcal{L}_C(W)=0$ implies that $\mathcal{L}_{DAG}(W)+\lambda_{CI}\|M\circ W\|_F^2$ is zero, and since $\sum_{p=1}^{d}C_p tr((W\circ W)^p)$ and $\|M\circ W\|_F^2$ are both non-negative, so $\sum_{p=1}^{d}C_p tr((W\circ W)^p)$ and $\|M\circ W\|_F^2$ must be all zero. Since a DAG does not have any self-loop, the diagonal entries of W and M must be zero, and so are the diagonal entries of $\nabla\mathcal{L}_C(W)$. Now, we need to consider the (i,j)th entry of $\mathcal{L}_C(W)$ with $i\neq j$:

- If there is no directed edge from node i to node j, i.e. $W_{ij}=0$, then it is clear that $\nabla\mathcal{L}_C(W)=2*(\sum_{p=1}^{d}C_p\cdot p\cdot((W\circ W)^{p-1^T})_{ji}+M_{ij}^2)\circ W_{ij}=0$.
- If there is an edge from node i to node j in a graph, represented by a non-zero weight W_{ij}, i.e. $W_{ij}\neq 0$, then the corresponding entry in other term $\sum_{p=1}^{d}C_p\cdot p\cdot((W\circ W)^{p-1^T})_{ji}$ and M_{ij}^2 must be zero. If $\sum_{p=1}^{d}C_p\cdot p\cdot((W\circ W)^{p-1^T})_{ji}$ is non-zero, it implies that there is a direct edge from node j to node i, which contradicts the statement that W represents DAG. If M_{ij}^2 is non-zero, the edge W_{ij} does not exist, which contradicts the statement that there is a non-zero weight edge W_{ij} from node i to node j in the graph, hence $((W\circ W)^{p-1^T})_{ji}$ and M_{ij}^2 must be zero.

Therefore, at least one of W_{ij} and $\sum_{p=1}^{d}C_p\cdot p\cdot((W\circ W)^{p-1^T})_{ji}+M_{ij}^2$ must be zero. Combining the above cases, we have $\nabla\mathcal{L}_C(W)=0$. Another direct consequence is that embedding the WIC constraint into continuous optimization methods that use DAG acyclicity constraint $\mathcal{L}_{DAG}(W)=tr(e^{(W\circ W)})-d$ (e.g. NOTEARs and GOLEM) or $\mathcal{L}_{DAG}(W)=tr((I+\mu W\circ W)^d)-d$ (e.g. DAG-GNN) naturally is guaranteed to converge.

We then complete this proof. □

Appendix C. Proof of Theorem 2

Theorem 2. *If the underlying DAG satisfies Assumptions 3 and 4, then the maximum likelihood estimator with DAG penalty and CI regularization penalty asymptotically outputs a DAG that is quasi equivalent to the ground truth DAG and has the same number of edges.*

Proof. Let G^* and Θ denote the ground truth DAG and the generated distribution. Let W and Ω denote the weighted adjacency matrix and the diagonal matrix containing noise variances. Considering the weights of the sparsity constraint and the regularization constraint in WIR, the likelihood term dominates asymptotically, we will find a pair $(\hat{W}, \hat{\Omega})$ to denote the directed graph corresponding to $\hat{W}$ by $\hat{G}$ and since $\Theta \in \Theta(\hat{G})$, we have $H(\hat{G}) \subseteq H(G^*)$ under the credibility assumption. Due to the WIR constraint generating a directed graph with the same number of edges, we have $|E(\hat{G})| \leq |E(G^*)|$, and we have $H(\hat{G}) \not\subset H(G^*)$ by Assumption 4. Now, from $H(\hat{G}) \subseteq H(G^*)$ and $H(\hat{G}) \not\subset H(G^*)$ we conclude that $H(\hat{G}) = H(G^*)$. Therefore, the WIR constraint can generate an estimated graph that is quasi equivalent to the ground truth DAG.

We then complete this proof. □

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